Logic Circuit & Design

Zape, Eduardo Jose S.

BSCpE 3 - 102

(Midterm)

03 Task Performance 01

7-Segment Display (5-Input) – Full Documentation

Oct 14, 2026 ·

1. Overview

This project drives one 7-segment display from five input switches so that rows 0–15 of the truth table spell the name EduardoJoseSZape and rows 16–31 show a set of extra letters and digits.

Item	Value
Inputs	a (MSB), b, c, d, e (LSB)
Input rows	32, where row = 16a + 8b + 4c + 2d + e
Outputs	Sa top, Sb upper right, Sc lower right, Sd bottom, Se lower left, Sf upper left, Sg middle
Output logic	Common cathode: 1 = segment on (invert all outputs for common anode)
Rows 0–15 (a = 0)	E d u a r d o J o s e S Z a p e
Rows 16–31 (a = 1)	H L c b h n t y U F 9 8 7 6 5 4
Method	Truth table, then one 5-variable K-map per segment, then a sum-of-products expression per segment

The segments are named Sa–Sg so they do not clash with the input names a–e. The decimal point is not used.

2. Truth table

The 32 rows below are the full specification. Each output column is 1 when that segment is lit.

Row	a b c d e	Char	Sa	Sb	Sc	Sd	Se	Sf	Sg
0	0 0 0 0 0	E	1	0	0	1	1	1	1
1	0 0 0 0 1	d	0	1	1	1	1	0	1
2	0 0 0 1 0	u	0	0	1	1	1	0	0
3	0 0 0 1 1	a	1	1	1	1	1	0	1
4	0 0 1 0 0	r	0	0	0	0	1	0	1
5	0 0 1 0 1	d	0	1	1	1	1	0	1
6	0 0 1 1 0	o	0	0	1	1	1	0	1
7	0 0 1 1 1	J	0	1	1	1	1	0	0
8	0 1 0 0 0	o	0	0	1	1	1	0	1
9	0 1 0 0 1	s	1	0	1	1	0	1	1
10	0 1 0 1 0	e	1	1	0	1	1	1	1
11	0 1 0 1 1	S	1	0	1	1	0	1	1
12	0 1 1 0 0	Z	1	1	0	1	1	0	1
13	0 1 1 0 1	a	1	1	1	1	1	0	1

Row	a b c d e	Char	Sa	Sb	Sc	Sd	Se	Sf	Sg
14	0 1 1 1 0	p	1	1	0	0	1	1	1
15	0 1 1 1 1	e	1	1	0	1	1	1	1
16	1 0 0 0 0	H	0	1	1	0	1	1	1
17	1 0 0 0 1	L	0	0	0	1	1	1	0
18	1 0 0 1 0	c	0	0	0	1	1	0	1
19	1 0 0 1 1	b	0	0	1	1	1	1	1
20	1 0 1 0 0	h	0	0	1	0	1	1	1
21	1 0 1 0 1	n	0	0	1	0	1	0	1
22	1 0 1 1 0	t	0	0	0	1	1	1	1
23	1 0 1 1 1	y	0	1	1	1	0	1	1
24	1 1 0 0 0	U	0	1	1	1	1	1	0
25	1 1 0 0 1	F	1	0	0	0	1	1	1
26	1 1 0 1 0	9	1	1	1	1	0	1	1
27	1 1 0 1 1	8	1	1	1	1	1	1	1
28	1 1 1 0 0	7	1	1	1	0	0	0	0
29	1 1 1 0 1	6	1	0	1	1	1	1	1

Row	a b c d e	Char	Sa	Sb	Sc	Sd	Se	Sf	Sg
30	1 1 1 1 0	5	1	0	1	1	0	1	1
31	1 1 1 1 1	4	0	1	1	0	0	1	1

3. Letters used

A 7-segment display can only approximate letters, so some characters share one pattern and cannot be told apart on the display.

Pattern (a b c d e f g)	Rows	Characters
1011011	9, 11, 30	s, S, 5
1101101	12	Z (same shape as the digit 2)
0011101	6, 8	o (lowercase, used twice)
0111101	1, 5	d (used twice)
1111101	3, 13	a (used twice)
1101111	10, 15	e (used twice)

Rows 0–15 spell the name in order: E d u a r d o J o s e S Z a p e (16 characters, one per row). Rows 16–31 are free choices: H L c b h n t y U F 9 8 7 6 5 4. Change any row in the simulator to see its pattern, then update the truth table and K-maps to match.

4. K-maps

A 5-variable K-map has 32 cells, so each segment needs two 4×4 layers: one for a = 0 (rows 0–15) and one for a = 1 (rows 16–31). Rows are bc and columns are de, both in Gray order (00, 01, 11, 10), so neighbors differ in one variable and the edges wrap around. A cell in one layer is also adjacent to the cell in the same position in the other layer, so a group can span both layers and drop a.

Segment Sa

a = 0

bc \ de	00	01	11	10
00	1	0	1	0
01	0	0	0	0
11	1	1	1	1

bc \ de	00	01	11	10
10	0	1	1	1

a = 1

bc \ de	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	0	1
10	0	1	1	1

Segment Sb

a = 0

bc \ de	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	1	1	1	1
10	0	0	0	1

a = 1

bc \ de	00	01	11	10
00	1	0	0	0
01	0	0	1	0
11	1	0	1	0
10	1	0	1	1

Segment Sc

a = 0

bc \ de	00	01	11	10
00	0	1	1	1
01	0	1	1	1
11	0	1	0	0

bc \ de	00	01	11	10
10	1	1	1	0

a = 1

bc \ de	00	01	11	10
00	1	0	1	0
01	1	1	1	0
11	1	1	1	1
10	1	0	1	1

Segment Sd

a = 0

bc \ de	00	01	11	10
00	1	1	1	1
01	0	1	1	1
11	1	1	1	0
10	1	1	1	1

a = 1

bc \ de	00	01	11	10
00	0	1	1	1
01	0	0	1	1
11	0	1	0	1
10	1	0	1	1

Segment Se

a = 0

bc \ de	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1

bc \ de	00	01	11	10
10	1	0	0	1

a = 1

bc \ de	00	01	11	10
00	1	1	1	1
01	1	1	0	1
11	0	1	0	0
10	1	1	1	0

Segment Sf

a = 0

bc \ de	00	01	11	10
00	1	0	0	0
01	0	0	0	0
11	0	0	1	1
10	0	1	1	1

a = 1

bc \ de	00	01	11	10
00	1	1	1	0
01	1	0	1	1
11	0	1	1	1
10	1	1	1	1

Segment Sg

a = 0

bc \ de	00	01	11	10
00	1	1	1	0
01	1	1	0	1
11	1	1	1	1

bc \ de	00	01	11	10
10	1	1	1	1

a = 1

bc \ de	00	01	11	10
00	1	0	1	1
01	1	1	1	1
11	0	1	1	1
10	0	1	1	1

5. Boolean expressions

Each segment is a minimized sum of products (SOP) generated with the Quine–McCluskey method. A prime (′) means NOT and letters written together mean AND. All 32 rows are specified, so there are no don't-care cells.

Segment	Minimized SOP	Product terms
Sa	bcd′ + bde′ + bea′ + bec′ + dea′c′ + a′b′c′d′e′	6
Sb	cde + bca′ + ea′b′ + abdc′ + bcd′e′ + bda′e′ + ac′d′e′	7
Sc	abd + ade + ced′ + ad′e′ + da′b′ + ea′c′ + ba′c′d′	7
Sd	db′ + dc′ + ea′ + a′c′ + ade′ + ba′d′ + bc′e′ + eb′c′ + bced′	9
Se	ca′ + a′b′ + a′e′ + b′e′ + aec′ + ced′ + c′d′e′	7
Sf	bd + abe + ade + bec′ + ac′d′ + acb′e′ + b′c′d′e′	7
Sg	bd + be + a′d′ + acb′ + dec′ + ab′e′ + ca′e′	7

To check an expression by hand, take any truth-table row, substitute its a b c d e values, and confirm the SOP gives that row's output bit.

6. Gate-level circuit

The circuit is a two-level AND–OR network: five shared input rails with their inverted copies, one AND gate per product term, and one OR gate per segment feeding the display.

Part	Count	Notes
Input switches	5	a, b, c, d, e
NOT gates	5	One per input, giving a′ b′ c′ d′ e′
AND gates	50	6 + 7 + 7 + 9 + 7 + 7 + 7 product terms for Sa–Sg, each with 2 to 5 inputs
OR gates	7	One per segment, with 6 to 9 inputs
Display	1	7-segment, common cathode